Ch 1.3

② col 1 − 2 col 2 = col 3 → 2D plane in R^3

③ $C = [B \quad -B] = \begin{bmatrix} 1 & 2 & -1 & -2 \\ 2 & 1 & -2 & -1 \\ 3 & 3 & -3 & -3 \end{bmatrix}$

↓ same col space as B

⑥ A has 2 independent col.

B " 3 "

A+B has 3 " "

rule: rank (A+B) ≤ rank (A) + rank B

$$A+B: \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 2 \\ 5 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}$$

⑦ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

A+B $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ → 2

$A+B$ [$\hookrightarrow$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

$A+B$ $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\rightarrow$ 0 rank

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \; B \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$A+B = I \rightarrow$ rank 4.

(9) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ more 7 ones.

(13) perpendicular to $(1,1,1)$

$(x,y,z).(1,1,1) = x+y+z=0$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -1 \\ -1 & 0 & -1 \end{bmatrix}$$

$\text{col } 3 = v_1 + v_2 - v_3$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 2 & 2 & 0 \\ 0 & -2 & -2 \\ -2 & 0 & 0 \end{bmatrix}$$

$\text{col } 3 = v_2 - v_1$

(15) if $Ax=b$ then extra col b in $[A\,b]$ is comb^n of the 1st cols. so rank same.

(16) a) If $m \times n$ Mat A has n indep cols, the $m \geq n$

cz: n ind cols could not be in space R^m with $m<n$ so $m \geq n$ Allahu Akbar!

(b) if entries r random & $m=n$ or $m \geq n$, the cols r surely indep. !

(19) $\begin{bmatrix} a \\ b \end{bmatrix} = m \begin{bmatrix} c \\ d \end{bmatrix}$ $\therefore$ $a = mc$

$b = md \rightarrow m = \frac{b}{d}$

$\therefore a = \frac{b}{d} \cdot c \therefore \frac{a}{b} = \frac{c}{d}$

$\therefore (a, c) = M(b, d)$

↓ ratio.

(23) $Ax=0 \rightarrow$ x is $\perp$ to each row of A.

(3 dot prods are 0). so x is $\perp$ to all combinations of rows. x is $\perp$ to row-space (a plane).

Every x in nullspace of A ($Ax=0$) is $\perp$ to every vector in row space